

LECTURE SHEET

EXERCISE

(Radioactivity : Isotopes, Isobars, Kinetics of radioactive decay, Carbon dating, Nuclear reaction and applications)

LEVEL-I (MAIN)

Straight Objective Type Questions

- Which one of the following nuclear reaction is correct :
 - ${}_6\text{C}^{13} + {}_1\text{H}^1 \rightarrow {}_7\text{N}^{13} + {}_{-1}\beta^0 + \gamma$
 - ${}_{11}\text{Na}^{23} + {}_1\text{H}^1 \rightarrow {}_{10}\text{Ne}^{20} + {}_2\text{He}^4$
 - ${}_{13}\text{Al}^{23} + {}_0\text{n}^1 \rightarrow {}_{11}\text{Na}^{23} + {}_{-1}\text{e}^0$
 - ${}_{12}\text{Mg}^{24} + {}_2\text{He}^4 \rightarrow {}_{13}\text{Al}^{27} + {}_0\text{n}^1$
- Decrease in atomic no. is observed during :
 - Alpha emission
 - Electron capture
 - Positron emission
 - all
- The number of α - and β - particles emitted during the transformation of ${}_{90}^{232}\text{Th}$ to ${}_{82}^{208}\text{Pb}$ is respectively.
 - 2, 2
 - 4, 2
 - 6, 4
 - 8, 6
- The number of β - particle emitted during the change ${}_a\text{X}^c \rightarrow {}_d\text{Y}^b$ is :
 - $\frac{a-b}{4}$
 - $d + \left[\frac{a-b}{2} \right]$
 - $d - a + \left[\frac{c-b}{2} \right]$
 - $d + \left[\frac{a-b}{2} \right] - c$
- The triad of nuclei that represents isotopes is :
 - ${}_6\text{C}^{14}, {}_7\text{N}^{14}, {}_9\text{F}^{19}$
 - ${}_6\text{C}^{12}, {}_7\text{N}^{14}, {}_9\text{F}^{19}$
 - ${}_6\text{C}^{14}, {}_6\text{C}^{13}, {}_6\text{C}^{12}$
 - ${}_6\text{C}^{14}, {}_7\text{N}^{14}, {}_9\text{F}^{17}$
- Which of the following nuclei is unstable?
 - ${}_5\text{B}^{10}$
 - ${}_4\text{Be}^{10}$
 - ${}_7\text{N}^{14}$
 - ${}_8\text{O}^{16}$
- Neutrons are more effective projectiles than protons because they
 - Are attracted by the nuclei
 - Are not repelled by the nuclei
 - Travel with high speed
 - None of above
- The end product of $(4n + 2)$ disintegration series is
 - ${}_{82}\text{Pb}^{204}$
 - ${}_{82}\text{Pb}^{208}$
 - ${}_{82}\text{Pb}^{206}$
 - ${}_{82}\text{Pb}^{209}$
- The negative value of packing fraction indicates that the isotope is
 - Unstable
 - Very stable
 - Artificial
 - Stable
- The number of protons and neutrons for most stable element is
 - Even-odd
 - Even-even
 - Odd-odd
 - Odd-even
- If a radioactive element is placed in an evacuated container its rate of disintegration
 - will be increased
 - will be decreased
 - will change very slightly
 - will remain unchanged
- C-14 has a life of 5760 years. 100 mg of sample containing C-14 is reduced to 25 mg in
 - 11520 years
 - 2880 years
 - 1440 years
 - 17280 years

13. One curie of activity is equivalent to
 1) 3.7×10^7 disintegrations per second
 2) 3.7×10^{10} disintegrations per second
 3) 3.7×10^4 disintegrations per second
 4) None
14. The activity of a radionuclide (X^{100}) is 6.023 curie. If the disintegration constant is $3.7 \times 10^4 \text{ sec}^{-1}$, the mass of radionuclide is :
 1) 10^{-14} g
 2) 10^{-6} g
 3) 10^{-15} g
 4) 10^{-3} g
15. A radioactive substance has a half-life of 50 days. Fraction of the material left behind after 100 days will be :
 1) 50%
 2) 75%
 3) 12.5%
 4) 25%
16. The equipment used to carry out nuclear reaction in a controlled manner is called
 1) Breeder reactor
 2) Nuclear reactor
 3) Thermonuclear fission
 4) Cyclotron
17. Which of the following is not a fissionable material ?
 1) U^{238}
 2) U^{234}
 3) Pu^{239}
 4) U^{235}
18. If two light nuclei are fused together in nuclear reaction, the average energy per nucleon
 1) Increases
 2) Decreases
 3) Cannot be determined
 4) Remains same
19. In nuclear reactors heavy water is used as
 1) Fuel
 2) Projectile
 3) moderator
 4) Arrestor
20. The radioisotope used in the treatment of cancer is
 1) C - 12
 2) Co-60
 3) I^{131}
 4) P-31
21. Which of the radioactive isotopes is used for temperature control in blood disease?
 1) P^{32}_{15}
 2) H^3
 3) Rn^{222}
 4) I^{131}

Numerical Value Type Questions

22. In the sequence of the following nuclear reaction $^{238}_{98}\text{X} \xrightarrow{-\alpha} \text{Y} \xrightarrow{-\beta} \text{Z} \xrightarrow{-\beta} \text{L} \xrightarrow{n\alpha} ^{218}_{90}\text{M}$
 What is the value of n
23. The $^{235}_{92}\text{U}$ disintegrates to give 4α and 6β particles. The atomic number of the end product is
24. The binding energy of an element is 64 MeV. If BE/nucleon is 6.4, the number of nucleons are :
25. A radioisotope has half life of 10 years. what percentage of the original amount of it would you expect to remain after 20 years?
26. If 75% quantity of a radioactive isotope disintegrates in 2 hour, its half life would be _____ minute.

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. Atomic weight of Th is 232 and its atomic number is 90. The number of α - and β - particles which will be lost so that an isotope of lead (atomic weight 208 and atomic number 82) is produced is
 a) $4\alpha + 6\beta$
 b) $6\alpha + 4\beta$
 c) $8\alpha + 2\beta$
 d) $10\alpha + 2\beta$
2. In which of the following the magic numbers of both protons and neutrons are present
 a) Sn^{123}_{50}
 b) Pb^{208}_{82}
 c) Pb^{206}_{82}
 d) Sn^{118}_{50}

3. The half-life of I^{131} is 8 days. Given a sample of I^{131} at $t = 0$, we can assert that :
- No nucleus will decay at $t = 4$ days
 - No nucleus will decay before $t = 8$ days
 - All nucleus will decay before $t = 16$ days
 - A given nucleus may decay at any time after ($t = 0$)
4. A wooden article and a freshly cut tree show activity $7.6 \text{ counts min}^{-1} \text{g}^{-1}$ and $15.2 \text{ counts min}^{-1} \text{g}^{-1}$ of carbon ($t_{1/2} = 5760 \text{ year}$) respectively. The age of the article is :
- 5760 year
 - $5760 \times \frac{15.2}{7.6} \text{ year}$
 - $5760 \times \frac{7.6}{15.2} \text{ year}$
 - $5760 \times (15.2 - 7.6) \text{ year}$
5. The radioactive decay rate of a radioactive element is found to be 10^3 dps at a certain time. If the half-life of element is 1 sec, the decay rate after 1 sec, is _____ and after 3 sec, is _____
- 500, 125
 - 125, 500
 - $10^3, 10^3$
 - 100, 10
6. The counting rate observed from a radioactive source at $t = 0$ seconds was 1600 counts/sec and at $t = 8 \text{ sec}$ it was 100 counts/sec. The counting rate observed as count per sec at $t = 6 \text{ sec}$ will be
- 400
 - 300
 - 200
 - 150
7. The number of millimoles of ${}^{14}_6\text{C}$ equivalent to one millicurie, if $t_{1/2} = 5570 \text{ year}$ and $1 \text{ curie} = 3.7 \times 10^{10} \text{ dps}$ is
- 1.56×10^{-2}
 - 3.12×10^{-2}
 - 4.34×10^{-2}
 - 7.80×10^{-2}
8. If N_0 is the initial number of nuclei, number of nuclei remaining undecayed at the end of n th half-life is :
- $2^{-n}N_0$
 - 2^nN_0
 - $n^{-2}N_0$
 - n^2N_0
9. The $4n$ series starts from Th-232 and ends at
- Pb-208
 - Bi-209
 - Pb-206
 - Pb-207
10. Living things contain C^{12} and C^{13} , C^{12} is stable and C^{13} decays and declines in proportional quantity. The technique that used this principle for determining the age of fossils skeletons, old trees, and dinosaurs is called
- C-12 dating
 - Radiocarbon dating
 - Carbon age
 - Fossil carbon
11. A radioactive substance (parent) decays to its daughter element, the age of radioactive substance (t) is related to the daughter (d) parent (p) ratio by the equation :
- $t = \frac{1}{\lambda} \ln \left(1 + \frac{p}{d} \right)$
 - $t = \frac{1}{\lambda} \ln \left(1 + \frac{d}{p} \right)$
 - $t = \frac{1}{\lambda} \ln \left(\frac{d}{p} \right)$
 - $t = \frac{1}{\lambda} \ln \left(\frac{p}{d} \right)$
12. The decay of a radioactive element follows first order kinetics. Thus,
- Half-life period = a constant/ K , where K is decay constant
 - The rate of decay is independent of temperature
 - The rate can be altered by changing chemical conditions
 - The element will be completely transformed into new element after expiry of two half-life period
13. Slow neutrons can bring about the fission of
- ${}^{235}_{92}\text{U}$
 - ${}^{238}_{82}\text{U}$
 - ${}^{207}_{82}\text{Pb}$
 - ${}^{226}_{88}\text{U}$

14. Which of the following pairs are isodiapheric pairs ?

- a) ${}_{29}\text{Cu}^{65}$ and ${}_{24}\text{Cr}^{55}$ b) ${}_{29}\text{Cu}^{65}$ and ${}_{24}\text{Cr}^{52}$ c) ${}_{92}\text{U}^{235}$ and ${}_{90}\text{Th}^{231}$ d) ${}_{92}\text{U}^{238}$ and ${}_{90}\text{Th}^{231}$

More than One correct answer Type Questions

15. If the densities of nuclei of ${}_1\text{H}^1$ and ${}_{92}\text{U}^{238}$ are X & Y respectively, then incorrect relation is

- a) $X = Y$ b) $X < Y$ c) $Y = 92X$ d) $Y = 238 X$

16. Which among these is/are correctly matched ?

- a) Positron emission : n/p ratio increase b) K - electron capture : n/p decreases
c) β^- - decay : n/p ratio decreases d) α - decay : n/p ratio increases

17. Which of the following are doubly magic?

- a) ${}_2\text{He}^4$ b) ${}_8\text{O}^{16}$ c) ${}_{82}\text{Pb}^{208}$ d) ${}_{92}\text{U}^{238}$

18. Decrease in atomic number is observed during :

- a) α - emission b) β^- - emission c) positron emission d) electron capture

19. Which of the following are α - emitters?

- a) Po^{213} b) Pb^{215} c) Rn^{222} d) Ra^{226}

20. Which of the following nuclides belong to actinium (U^{235}) series?

- a) Pb^{207} b) Po^{215} c) Po^{213} d) ${}_1\text{H}^3$

21. Which among the following nuclides is/are likely to be stable?

- a) ${}_{49}\text{In}^{114}$ b) ${}_{12}\text{Mg}^{24}$ c) ${}_{48}\text{Cd}^{114}$ d) ${}_{15}\text{P}^{30}$

22. Which of the following is/are true?

- a) The most of radioactive element present in pitchblende is uranium
b) P^{32} is use for the treatment of leukaemia
c) CO_2 present in the air contains C-12 only
d) Emission of γ -rays changes the mass number but not atomic number

Linked Comprehension Type Questions

Passage-I :

The emission of penetrating α, β - particles (${}_2^4\text{He}$ and ${}_{-1}^0\text{e}$ respectively) along with γ -radiation ($h\nu$) was noticed from unstable nuclei. All elements having $Z > 82$ show this phenomenon. The emission was explained in terms of low binding energy (giving α -decay), high n/p ratio (neutron decay). γ -emission from a radioactive nuclide is secondary emission.

23. An element of group III with At. no. 90 and mass number 238 undergoes decay of one α -particle. The newly formed element belongs to :

- a) I b) II c) III d) IV

24. The emission of penetrating rays from a radioactive species can be shielded by :

- a) Bi blocks b) Pb blocks c) C blocks d) Mg blocks

25. The binding energy of the hydrogen nuclide is :

- a) zero b) 13.6 eV c) > 13.6 eV d) < 13.6 eV

Passage-II :

It has been estimated that the total power radiated by the sun is 3.8×10^{26} J per second. The source of energy of stars is a thermonuclear fusion reaction. Energy released in the process of fusion is due to mass defect. $Q = \Delta mc^2$

26. B.E. per nucleon of ${}^2_1\text{H}$ and ${}^4_2\text{He}$ are 1.1 MeV and 7 MeV respectively. If two deuteron nuclei react to form a single helium nucleus, then the energy released is :
 a) 13.9 Mev b) 26.9 Mev c) 23.6 Mev d) 19.2 Mev
27. Mass equivalent to energy 931 Mev is :
 a) 6.023×10^{-27} kg b) 1.66×10^{-27} kg
 c) 16.66×10^{-27} kg d) 16.02×10^{-27} kg
28. In a nuclear reaction, ${}^2_1\text{H} + {}^2_1\text{H} \rightarrow {}^3_2\text{He} + {}^1_0\text{n}$ if the masses of ${}^2_1\text{H}$ & ${}^3_2\text{He}$ are 2.014741 amu and 3.016977 amu respectively, then the Q-value of the reaction is nearly :
 a) 0.00352 Mev b) 3.63 Mev c) 0.82 Mev d) 2.45 Mev

Passage-III :

Unstable nuclei attain stability through disintegration. The nuclear stability is related to neutron proton ratio (n/p). For stable nuclei n/p ratio lies close to unity for elements with low atomic numbers (20 or less) but it is more than 1 for nuclei having higher atomic numbers. Nuclei havign n/p ratio either very high or low undergo nuclear transformation. When n/p ratio is higher than requird for stability, the nuclei have the tendency to emit β -rays. Which when n/p ratio is lower than required for stability, the nuclei either emits α -particles or a positron or capture K-electron.

29. Unstable substance exhibit high radioactivity due to
 a) Low p/n ratio b) high p/n ratio c) p/n = 1 d) None
30. β -particle is emitted in radioactivity by
 a) Conversion of proton to neutron
 b) Conversion of neutrons to proton
 c) β - particle is not emitted
 d) None
31. Among the following nuclides, the highest tendency to decay by β^+ emission is
 a) Cu^{68} b) Cu^{59} c) Cu^{69} d) Cu^{67}

Passage-IV :

In nuclear reactor, moderator is used to slow down the neutrons produced during nuclear fission. Heavy water or graphite moderators slow down the speed of neutrons. The essential characteristics of a moderator are :

- Its molar mass must be low
- It should not absorb neutrons
- It should undergo elastic collision with neutrons

32. The moderator in a reactor
 a) Absorbs neutrons
 b) Accelerate neutrons
 c) Slows down neutrons
 d) Absorbs thermal energy produced in the reactors
33. Which of the following is not used as the moderator ?
 a) Heavy water
 b) Graphite
 c) Beryllium
 d) Sodium
34. Moderator in the reactor yields
 a) Fast moving neutron
 b) Thermal neutron
 c) Magnetic neutron
 d) Electric neutron

Matrix Matching Type Questions35. **Column-I**

- A) Proton rich nuclides
 B) Artificially prepared element
 C) ${}^{11}_6\text{C}$
 D) C - N cycle

Column-II

- p) K-electron capture
 q) Proton emission
 r) Positron emission
 s) ${}^{97}_{43}\text{Tc}$

36. **Column-I (Particles emitted)**

- A) One α -particle
 B) One α - and two β - particles
 C) One β - particle
 D) γ -radiation

Column-II (Result)

- p) Isobar
 q) Isotope
 r) Isodiapher
 s) Atomic number reduced by two and mass number by four
 t) Nuclear de-excitation

37. **Column-I**

- A) ${}^{32}\text{P}$
 B) ${}^{24}\text{Na}$
 C) ${}^{60}\text{Co}$
 D) ${}^{131}\text{I}$

Column-II (Characteristic Use)

- p) Location of tumour in brain
 q) Location of blood clot and circulatory disorders
 r) Radiotherapy
 s) Agriculture research

Integer Type Questions

38. The number of following unstable nucleides emitted β -particle. ${}_{20}\text{Ca}^{40}$, ${}_{53}\text{I}^{133}$, ${}_{53}\text{I}^{121}$, ${}_{90}\text{Th}^{232}$, ${}_{6}\text{C}^{14}$, ${}_{7}\text{N}^{13}$, ${}_{92}\text{U}^{235}$
39. The halflife of C-14 is 5600 years. A sample of freshly cut wood from a tree contains 10 mg of C-14. The amount left in the sample after 50000 years is $(a - x)$. The value of $(a - x) \times 100$ is
40. In the following given numbers, how many are the magic numbers ?
 2, 8, 20, 50, 113, 126, 141, 148, 300, 314

OBJECTIVE CHEMISTRY IID**NUCLEAR CHEMISTRY****KEY SHEET (LECTURE SHEET)****EXERCISE****LEVEL-I**

- 1) 2 2) 4 3) 3 4) 3 5) 3 6) 2 7) 2 8) 3
9) 4 10) 2 11) 4 12) 1 13) 2 14) 3 15) 4 16) 2
17) 1 18) 2 19) 3 20) 2 21) 1 22) 4 23) 90 24) 10
25) 25 26) 60

LEVEL-II

- 1) b 2) b 3) d 4) a 5) a 6) c 7) a 8) a
9) a 10) b 11) b 12) a 13) a 14) c 15) bcd 16) acd
17) abc 18) acd 19) ad 20) ab 21) bc 22) bc 23) b 24) b
25) a 26) c 27) b 28) b 29) a 30) b 31) d 32) c
33) b 34) a 35) A-pqr; B-s; C-r; D-r 36) A-rs; B-q; C-p; D-t
37) A-s; B-q; C-r; D-p 38) 3 39) 2 40) 5